Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Paper 5 Memory Rank Review Response

Benjamin D. Summers <bsummers@adagiollc.com> Thu, Sep 17, 2026 at 6:01 PM
To: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>, Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Hi Ezequiel and Sara,

Thank you both for your reviews (both attached). The revised manuscript has been uploaded to SSRN (https://dx.doi.org/10.2139/ssrn.7119779). This time you did not review the same text. Sara's review quotes the version deposited in July at every point she cites (0.062434, 0.010136, 0.089160, κ_c = 0.276613, the floor 0.195595, the block masses 10^{-23}, 10^{-54}, 10^{-27}, Definition 5.9, the homogenized history lift). Ezequiel's report cites the same SSRN DOI on its cover but quotes the earlier text at every checkable point: the abstract's "extend to a flat transport on a d-state structured enlargement", "the categorical quotient collapses", the bounds $h_3 \le 0.026680$ and $h_5 \le 0.000222$, block masses of 10^{-13}, the constant C* = n(min p·π)^{-1/2}, "κ_c is never computed", and the Moreau sentence in Section 6. None of these is in the July version, so the report examined a superseded file. Findings 2 through 7, both local errors (the Moreau decomposition in Section 6 and the bound of Proposition 7.1), and several of the [R] items were addressed in the July revision, and I say where below. The two reviews still converge on the same points: the reading of a vanishing profile as approximability rather than realization, the one-sided character of the marginal-consistency obstruction, the standing of the compactness, stability and semialgebraicity statements relative to standard facts, the reproducibility of Section 8, and the name of the invariant. My responses follow in manuscript order, starting with the one that explains most of the others. Where you both raised a point, I answer it once. Section and result numbers are those of the revision; where they differ from the version you read I give both. Sara, your comments are not numbered, so I cite them by their lead phrases.

Ezequiel, thank you for the verification record. I checked every figure in your report that the printed definitions determine. The algebra agrees: the inverse and the conjugation formula of Theorem 3.3 with the entrywise bound, the strong-lumpability check behind Proposition 4.4, the constant √n, and D_max = 62 for n = 2 and m = 4. Your arithmetic agrees to the printed digit: 0.324012/2 = 0.162 with the July constant C* = n, and 1 − 0.910255/1.319266 = 0.3100 is your thirty-one percent on the July table. Your agreement to 2.2×10^{-16} between the four-state and three-state families in the split-and-lump construction is the rounding of an exact equality, which Proposition 4.4 gives. Your remaining figures are draws from families you did not print, and I do not set mine beside them, because the revision withdraws the numerical section (see Section 8 below). I must correct one item in your record. Your Section 3.1 says that Remark 2.3's reading of (5) as δ_VΔ_V = 0 "for the displayed twisted coboundary is correct". The display you read had the two twisted terms with the wrong sign: with that display δ_VΔ_V reduces to twice a difference of two deficits and not to zero. The July revision corrected the ordering, and the corrected display satisfies δ_VΔ_V = 0 identically; the revision adds the formula for δ_V^2 (Section 2 below). Your D_max argument, your joint-law obstruction, your witness, your split-and-lump construction, your identification of the re-dressing group, and the question that Proposition 4.11 answers are in the paper with attribution.

Your repair of the final clause of Theorem 5.7 explains most of the other changes. The July revision had deleted the false clause and posed termination on the closure as an open question (Definition 5.9, Open Problem (vii)(b)). Your argument proves it. Every process-induced family lies in $\kappa(\Theta_D^+)$ with $D \le D_{max} = \Sigma_{k=0}^{m} n^{k+1}$, so in $F_{D_{max}}$ by the exact inclusions of Proposition 4.4; $F_{D_{max}}$ is closed, so the closure of the process-induced families lies in it; and part (b) of the theorem puts every F_d inside that closure. So $F_d = F_{D_{max}}$ for every $d \ge D_{max}$, the union of the model sets is closed and equals the closure of the process-induced families, the profile of every family is constant from D_max on and equals its distance to that closure, and the memory rank is finite exactly on the closure, where it is at most D_max. This is Theorem 5.7(d). Definition 5.9 has become Corollary 5.9: the asymptotic liftability class is empty, in either class. Four other things follow. Open Problem (vii) keeps the compactification question and drops the termination question. The threshold rule of Remark 4.7 covers the infinite-rank case, because $h_d \ge h_{D_{max}} > 0$ for every d off the closure. Sara's objection that a compactification would not by itself close the union is correct and is now moot, since the union is closed without one. The abstract states the stabilization as a theorem in place of the open question.

Abstract, significance statement and introduction (Sara: "without a generator", the quantum analogy, the abstract's "attains zero exactly", novelty positioning, early qualification of "memory rank", "exactly" flat transports, the zero stratum as definitional, "computable"; Ezequiel Finding 8, Questions 13, 14 and 15, and the originality section). Accepted. "Without a generator" now means, in the abstract, the significance statement and the first paragraph, no time-local Markov generator on the observed state space; the dilations of Section 4 are generators on a hidden space, and the text says so. The quantum non-Markovianity connection is stated as a formal analogy, with the correspondence (column-stochastic kernels for completely positive maps, the deficit for the divisibility defect) and nothing more. The abstract no longer refers to computations at all; its last sentence lists the exact examples of Section 8. The abstract's first sentence now presents flatness as terminology (we call a family flat when it satisfies Chapman–

Kolmogorov and read flatness as a flat transport), and the first paragraph says "we read (1) geometrically". The abstract no longer presents the zero-stratum equivalence as a result; it states the stabilization and the two obstructions. The memory rank carries both qualifiers, closed-model and class-relative, at its first mention in the introduction, and the closed-model qualifier in the significance statement. "Dilation category" is "dilation class" throughout, Section 7 is "Class dependence", and no morphisms are claimed. The invariant is "computable in principle and bounded from below by the linear programs of Section 5". The contributions now come in two registers. The standard consequences of the definitions, stated because the companion papers use them, are compactness and attainment from the closure, stability as the 1-Lipschitz property of every distance function, semialgebraicity as Tarski–Seidenberg, and the first-order conditions of Section 6 as the general necessary conditions of variational analysis. The paper's own results are the gauge trivialization theorem with its corollary; the stabilization at D_max with the finiteness of the rank on the closure; the two linear-programming obstructions with their Lipschitz constants; the characterization of the closure gap with its intrinsic witness; grid monotonicity; and class dependence with an explicit gap. A positioning paragraph carries Ezequiel's four references and Sara's six areas: Finesso, Grassi and Spreij (2010) minimize an information divergence rate on the full process over hidden Markov models of assigned order, which is the shape of the profile; the profile instead measures a Euclidean distance on two-time conditionals to a model set indexed by hidden dimension on a finite grid, and adds the realization statements of Theorem 5.7 and the two obstructions. That answers Question 14. Casanellas, Fernández-Sánchez and Garrote-López (2021) compute distances to the stochastic part of a model variety, of which h_d is the temporal instance; Kubjas, Robeva and Sturmfels (2015) with Cohen and Rothblum (1993) give the one-time analog of the boundary of F_d. Balle, Lacroce, Panangaden, Precup and Rabusseau (2021) solve the linear one-letter case exactly by Adamyan–Arov–Krein theory because the model set there is a linear-algebraic object; positivity of the lift makes F_d semialgebraic and nonconvex, and extension complexity describes that obstruction. That answers Question 15. Heller (1965), Anderson (1999), Vidyasagar (2011), Ito, Amari and Kobayashi (1992), Critch (2013) and Gassiat and Boucheron (2003) cover realization, identifiability, the algebraic statistics of hidden Markov models and order estimation; Vorob'ev (1962), Kellerer (1964) and Arnold, Castillo and Sarabia (1999) cover the marginal problem and the compatibility of conditionals; the joint-law defect is a finite instance of that problem. Question 13 is Remark 5.12: the memory rank is the least closed-model hidden dimension reproducing the two-time conditionals; a realization of the path law by a function of a Markov chain is a lift, so the rank is at most the minimal order of such a realization of any inducing process, and at most n times the minimal order of a hidden Markov model with stochastic emissions, which becomes such a realization on state–symbol pairs; both inequalities can be strict; no inequality with the Hankel rank of a path law or the nonnegative rank of its two-time joint matrices is claimed in either direction, and Open Problem (iii) now asks for those comparisons on like data. I have no counterargument to Ezequiel's assessment that the individual results are low to moderate and that the value is in the architecture and the negative theorems; the two-register paragraph says so.

Section 2, the deficit (Sara: cochain and complex terminology, Remark 2.3, "deficit complex"; Ezequiel's [R] item on the higher coherence, Question 16). Accepted, and the answer to Question 16 is no. Define the V-twisted coboundary on 1-cochains by $(\delta_V\Psi)(t,s,r) = V(t\leftarrow r)\Psi(r,s) - \Psi(t,s) + \Psi(t,r)V(r\leftarrow s)$ and on 2-cochains by the corrected display. Then for every 1-cochain Ψ, $(\delta_V^2\Psi)(t,s,r,q) = \Psi(t,q)\Delta_V(q,s,r) - \Delta_V(t,r,q)\Psi(r,s)$, which is equation (6). So δ_V^2 vanishes when V is flat and is otherwise the action of the deficit on 1-cochains, and the pair (5)–(6) is the exact analog of $d_A F_A = 0$ with $d_A^2 = F_A$. That is the whole cohomological content, it is stated as such, and the associahedral speculation is gone; the section title lost the word "complex".

Section 3, the gauge route (Sara: the re-dressing class as "nonphysical", the GIT language, the action on the affine space; Ezequiel Finding 1 (i)–(iii), Question 12). Accepted, with one rebuttal. Question 12: the paper proves Corollary 3.4 and nothing more; Remark 3.5 now calls the picture an analogy with orbit degeneration and claims no instability, quotient or ring of invariants. Definition 3.1 says the re-dressings act on the affine space of unit-column-sum families and that admissibility selects the portion of an orbit inside the stochastic domain. The conclusion of the section is stated relative to the declared class. The rebuttal concerns "nonphysical". Remark 3.6, which is Ezequiel's item (iii), identifies the re-dressings with the similarity transformations relating quasi-realizations in hidden-Markov realization theory, with admissibility as the cone condition that positive realization theory imposes; the class is a recognized object of the framework the paper cites. The remark also records the two extremes Sara's comment points to: a re-dressing that maps the stochastic domain into itself together with its inverse is a time-dependent relabeling (applied to the reset families $e_b\mathbb{1}^\intercal$ it must be entrywise nonnegative, and a stochastic matrix with a stochastic inverse is a permutation matrix), under which the deficit norm with uniform weights is invariant, so the orbital minimum over that subclass is the deficit norm itself; the full class collapses every orbit onto the flat families. The intermediate class with a conditioning bound κ is dealt with by a proof: for a non-flat family the constrained infimum F_κ is attained and positive, while along the path of Theorem 3.3 it tends to zero on the same orbit, so F_κ is not constant on orbits, and Example 8.5 gives the rate at which it collapses as κ grows. How the trivialization interacts with the non-uniqueness results of positive realization theory is recorded as a question and not answered.

Section 4, the profile (Sara: weights "by the law of the process", the notation π_i, the finite union over labelings, Proposition 4.4's proof, $\mathbb{1}^\intercal R = \mathbb{1}^\intercal$, Remark 4.7, the title of Lemma 4.8, standard versus model-specific facts; Ezequiel "stability is not a rank test", Question 17). Accepted. The weights are positive reference weights ω_i, with process marginals as a choice when a process exists. Θ_d^+ is written as the finite disjoint union over labelings, and F_d as the union of the closures. The proof of Proposition 4.4 states P, R, PR = I_d, $\mathbb{1}^\intercal R = \mathbb{1}^\intercal$ and $\mathbb{1}^\intercal P = \mathbb{1}^\intercal$ with dimensions before the calculation. Lemma 4.8 is "Weight independence of the zero sets and the rank". The garbled clause in the proof of Proposition 4.5 was repaired in July; it now reads that h_d is 1-Lipschitz by Proposition 4.6, so h_d^2 is continuous. Remark 4.7 keeps the transfer of norm consistency and drops "estimation theorem in embryo"; it adds the sentence

you both asked for and the rule that makes it precise. The plug-in rank without a threshold satisfies liminf rk(V̂_N) ≥ rk(V), because the model sets are closed and nested, with strict inequality whenever V̂_N leaves F_r, so it is biased upward. The rule d̂_N = min{d: h_d(V̂_N) ≤ τ_N} with τ_N → 0 and τ_N/ε_N → ∞, where ε_N bounds dist(V̂_N, V), is consistent: for a finite rank r, h_r(V̂_N) ≤ ε_N ≤ τ_N and h_d(V̂_N) ≥ h_{r−1}(V) − ε_N > τ_N below r for all large N; for infinite rank, h_d(V̂_N) ≥ h_{D_max}(V) − ε_N > τ_N at every d. The margin h_{r−1}(V) exists and is unknown, so the rate of the family estimator is the quantity the threshold must dominate. That answers Question 17, and it is the consistency of the integer under those hypotheses and no more.

Section 4.3, the closure gap (Ezequiel Finding 2, Questions 1 and 2; Sara: boundary contact as parameter-level evidence, the "closed-model" qualifier). Accepted, and the July revision had already taken the terminological repair: F_d is a closure, h_d = 0 means approximability, and the rank is a closed-model rank, in Definition 4.2, Definition 5.1, Remark 4.9 and the abstract; as you put it, the reading changed and the definition did not. The July text did not characterize the gap. Remark 4.9 characterizes it: V ∈ F_d lies outside κ(Θ_d⁺) if and only if every approximating sequence of positive lifts has a block mass tending to zero, by compactness of the parameters and continuity of κ where the masses are bounded below. Question 1: at d = n, in the inhomogeneous class, a flat family lies in κ(Θ_n⁺) if and only if some strictly positive initial law keeps every V(i←0)π strictly positive, so the gap at d = n is exactly the flat families with degenerate forced marginals, and in the homogeneous class the same holds for flat families with equal one-step kernels; for d > n, whether a family in the gap can admit strictly positive consistent marginals in the sense of Lemma 5.4 is open, and the paper says so. Your witness is Example 4.10, in both classes, at distance 2ε√(number of pairs) from the lift with block mass ε. Question 2: in the witness the limiting column on the vanished block is inherited from the direction of approach, and the general question of which columns are identifiable in the limit is the compactification problem of Open Problem (vii); the statistical meaning of h_d = 0 reached only through such a sequence is approximability, as the closed-model reading asserts. Sara's point that the July numerics established boundary contact only at the parameter level is accepted; those numerics are withdrawn, and Example 4.10 is the intrinsic statement.
Proposition 4.11, grid monotonicity (Ezequiel Question 18; Sara on the growth of the bound with the grid). Added. Restriction to a sub-grid with inherited weights is a coordinate projection, so 1-Lipschitz, and it maps lifts to lifts with initial law p_{i₀}(θ) and step matrices the products between retained times; so h_d(V|_{𝕋′}) ≤ h_d(V) and the rank cannot decrease under refinement, in the inhomogeneous class and in the homogeneous class for sub-grids with constant index gaps. Question 18: no, the rank cannot decrease under refinement; whether it stabilizes is Open Problem (vi), and the paragraph after the proposition says that D_max grows exponentially in the number of grid steps and that the paper supplies no uniform bound along a refining sequence.

Section 5.1, the marginal-consistency obstruction (Ezequiel Finding 3, Questions 7 and 8; Sara: the obstruction tests pairwise consistency only). Accepted, in two stages. The July revision removed the invalid inference (Theorem 5.6 "predicts" a floor for a family "asserted" to be process-free), computed κ_c for that random family, and sharpened the constant to √n(min p_ij ω_i(a))^{-1/2}. Question 8: the random family is withdrawn with the numerical section; the obstruction is now evaluated by hand on the family of Example 8.2, where κ_c = 1 and the floor is 1/√2 at every dimension. The hypothesis and the necessity-only status are stated after the proof and not inside the theorem. Definition 5.5 says the defect tests pairwise consistency of one-time marginals only. The second stage is Section 5.3.

Example 5.2 and Proposition 5.3 (Ezequiel's [R] item on positivity; Sara on "inhom"). Accepted. Example 5.2 says that a flat family with a zero row in some V(i←0) is not the compression of any lift in Θ_n⁺, that F_n^{inhom} = Flat survives because F_n is a closure and flat families with strictly positive one-step kernels are dense in Flat, and that F_n^{hom} is a proper closed subset of Flat as soon as m ≥ 2. Proposition 5.3's constants: L is explicit and the exponent and C are not constructive; the paragraph after the proof says so. A bound on the exponent in n and the grid would be worth more than the statement, as you say; I have none.

Section 5.2, realization (Sara: "sufficiency", the history lift as standard state augmentation, the hidden clock, the crude bound, D depending on the realizing process, Definition 5.9's two questions; Ezequiel's [R] item on the closure clause). Accepted, and this is the D_max change above. The subsection is "Realization of process-induced families". The text before the theorem says that the history lift is the standard augmentation by the observed history and that the content is the dimension bookkeeping and the stabilization it yields. Equation (9), before the theorem, defines D_max, and the paragraph after the theorem says that D depends on the realizing process while D_max depends on n and m only, that D_max is exponential in the grid and crude, and that for n = 2 and m = 4, D_max = 62, while a lumped 4-state chain on that grid can have memory rank at most 3 (Example 8.1). Remark 5.8 says that the homogeneous realization contains a clock, that the classes differ in the minimal dimension and not in eventual liftability, and that the class dependence of Section 7 is a statement at fixed d.

Section 5.3, the joint-law defect (Ezequiel Finding 3 and Question 7; Sara: κ_c = 0 does not give a joint law). Accepted with a change of name and one qualification. The obstruction is Definition 5.10 and Theorem 5.11, written κ_J to keep it clear of the coboundary δ_V: κ_J(V) = min over joint laws q of max over i < j and a, b of |q_ij(a,b) − V(j←i){ba} q_i(a)|, a linear program with n^{m+1} + 1 variables. It vanishes on the process-induced families and on every F_d; it is Lipschitz with constant C′ = (min p_ij ω_i(a))^{-1/2}, so h_d ≥ κ_J/C′ for every d; κ_c ≤ n²κ_J, so κ_J detects everything κ_c detects; and V is process-induced exactly when the program attains zero at a law with positive one-time marginals, which a second linear program decides. Question 7: no to the first part, since κ_c = 0 does not imply a process, and yes to the second: κ_J is a strict strengthening as a detector, and Example 8.3 is a family with κ_c = 0 and κ_J = 1/6, both by hand. The qualification is about the floor. The two constants differ, C* = √n C′, and on a given family either floor may be the larger; in Example 8.2 they are 1/√2 and 1/4. So the paper says that the two

obstructions are ordered as detectors and not as floors, and that the larger floor applies. The closure has one consequence for the converse: κ_J vanishes on F{D_max}, which is the closure of the process-induced families, so the converse from κ_J = 0 to membership in the closure holds when some optimal law has positive marginals and is open when every optimal law is degenerate (Remark 5.12).

Section 6, the first-order conditions (Ezequiel's [M] item on Moreau, Question 10, the affine-hull [R] item; Sara: generic projection geometry, the residual is not the deficit, the ambient measure, the Moreau passage, chart stationarity versus stratum orthogonality, positive block masses versus interior). Accepted. Question 10 was answered in July: Moreau's theorem needs a closed convex cone, the contingent cone of a semialgebraic set need not be convex, the general statement is the polar-of-contingent-cone inclusion, and the decomposition applies against the Clarke tangent cone; the revision cuts that passage to one sentence. The section is retitled "First-order conditions at the projection", presented as the standard framework, and the number is kept because the companion paper cites it. The residual is called the model mismatch and the text says it is not the deficit. The null set is relative to Lebesgue measure on the affine hull, and the two senses of "generic" are kept apart. The one model-specific statement is Remark 6.1: the compression is invariant under the group of block-diagonal mass-preserving similarities, of dimension Σ_a d_a(d_a−1) per time in the inhomogeneous class and once in the homogeneous class, so at a point with discrete stabilizer the differential of the compression has rank at most the parameter dimension minus the group dimension, and at most the ambient dimension; I do not prove that the stabilizer is discrete generically. Sara's two points on the fitted points, that chart stationarity does not certify stratum orthogonality and that positive block masses do not place a point in the interior, are accepted, and the claims they concerned are withdrawn with the numerical section.

Section 7, class dependence (Ezequiel's [M] item on Proposition 7.1, Question 11; Sara: the hidden-clock qualification). Accepted. The error behind Question 11 was repaired in July: the two summands carry different weight vectors, so the proof passes to a common Frobenius lower bound with c = min ω_i(a), and the gap is (c min(p_01, p_12)/2)^{1/2}‖A − B‖_F. The strictness is stated at the dimension-matched level, and the paragraph after the proof says that at larger d the homogeneous class encodes time in a hidden clock, so the family of the proof lies in some F_D^{hom} with D ≤ D_max, and the profile at fixed d records the exchange rate; Example 8.4 computes it in closed form, h_2^hom = √3 against h_2^inhom = 0 for the family (I, S, S), where the bound of Proposition 7.1 gives √2.

Section 8 (Ezequiel Findings 5, 6 and 7, Questions 3, 4, 5, 6 and 9, the exchange-rate, "plateau" and arithmetic-floor items; Sara: the deposit, "certified upper bounds", the heading, "unambiguous", "projection", interior, chart stationarity, tiny block masses, the κ_c certificate, "plateau", the gauge computation, the determinant-floor experiment, labeling classes, parameter counts, "unprompted", "visibly", and the data statement). The numerical study is withdrawn from the paper: the tables, the optimizer, the fitted points, the seeds, the script, the certificates, and the data statement are gone. The reason is the one you both gave in different words. An optimizer returns upper bounds on h_d, and no upper bound establishes that h_{d−1} > 0; so the tables could not support the minimality of the rank, and the archive, environment and certificates they required served that claim. Section 8 now contains five exact examples, each proved by hand from the definitions. Example 8.1 is your split-and-lump construction in general form: the compression of a split lift is unchanged by Proposition 4.4, so the memory rank is at most the generating dimension of any lumped chain inducing the family, with strict inequality for every family induced by a split lift; that is my answer to Question 6. Example 8.2 is the family (R₁, R₁, R₂) of reset kernels on the three-point grid, with κ_c = 1 by a triangle inequality and κ_J = 1/4 by a short argument with the attaining law written down; its profile is at least 1/√2 at every dimension in either class, and the inequality κ_J ≥ κ_c/n² is attained. Example 8.3 is the family (I, I, S): uniform consistent marginals, so κ_c = 0, and κ_J = 1/6, since the events {X₀ = X₁}, {X₁ = X₂}, {X₀ ≠ X₂} cannot all hold; it is the smallest instance of the marginal problem of Vorob'ev and Kellerer in this setting, and it separates the two obstructions. Example 8.4 is the flat family (I, S, S): h_2^inhom = 0 and h_2^hom = √3 exactly, by the identification F_2^hom = {(M, M, M²)} and a closed-form expansion of the squared distance around the uniform reset. Example 8.5 is the gauge path in closed form: $\Delta_{V^g}(t,s,r) = \varepsilon^{t-s}\,\Delta_V(t,s,r)(I - (1-\varepsilon^{s+1})w\mathbb{1}^T)$, so every triple's deficit decays at the rate ε^{t−s}, the family approaches the reset form at the rate ε, the distance to the flat families decays like ε², and the bounded-conditioning surrogate of Remark 3.6 is O(κ^{−2/(m+1)}) in the conditioning bound κ. Question 3: an upper bound is consistent with zero, so no rank is claimed from computation; the lower bounds the paper states come from the two obstructions, evaluated exactly, and I produce no certificate of non-membership in F₃ for a realizable family, where both obstructions vanish. Questions 4 and 5: the table is gone; the zeros the paper asserts are exact (Examples 8.1 and 8.4) and the positive values are exact floors (Examples 8.2 and 8.3). Question 9: there is nothing to archive; the paper contains no computation. The exchange rate is Example 8.4, on a specified family, in closed form. "Plateau" is gone. The arithmetic floor of numerical inversion is not in the paper, since the gauge path is now a formula; your observation is right for any implementation that inverts g_i numerically, and the closed form of Example 8.5 avoids it. Sara's items on certification, headings, "projection", interior, stationarity, block masses, the κ_c certificate, the gauge computation, the determinant-floor experiment, labeling classes, parameter counts, "unprompted", "visibly", and the data statement are met by the withdrawal, and her request for a broader literature discussion by Section 1.

Section 9, open problems (Sara: non-unique projections in (i), h_n as a contraction in (ii), ΛV in (v), compactification versus termination; Ezequiel Question 2 and the [R] item on Open Problem (iii)). Accepted. Problem (i) is a design problem on the triple, quantified over all nearest points, with the equality clause on the zero stratum stated as automatic, in the form the Paper C revision uses. Problem (ii) asks for the contractions of Δ_V and says that h_n is not one. Problem (iii) compares the memory rank with the minimal hidden-Markov order, the Hankel rank and the nonnegative rank on the same data. Problem (v) defines ΛV on the process-induced families relative to the inducing process. Problem (vii) keeps the compactification question, with the witness saying what the boundary data must record and the d > n question of Remark 4.9.

Not adopted: Ezequiel's structural repair in Finding 2, a compactified class whose exact image is F_d, is posed and not carried out; Open Problem (vii) names Lemma 5.4 as the template, as you proposed. His statement that the memory rank equals the generating dimension for generic lumped chains is not printed: I believe it and cannot prove it. His "strict strengthening" holds as a detector and not as a floor, and the paper says which. Your shared request for a persistent public archive of the script, certificates and optimizer records is moot: the paper no longer contains a computation. I have no counterargument to Sara's recommendation of major revision for calibration and positioning or to Ezequiel's level assessment.

Bibliography: I added Finesso, Grassi and Spreij (MCSS 22, 2010), Casanellas, Fernández-Sánchez and Garrote-López (J. Symbolic Comput. 104, 2021), Balle, Lacroce, Panangaden, Precup and Rabusseau (ICALP 2021), Kubjas, Robeva and Sturmfels (Ann. Statist. 43, 2015), Critch (J. Algebraic Statistics 4, 2013), Ito, Amari and Kobayashi (IEEE Trans. Inform. Theory 38, 1992), Heller (Ann. Math. Statist. 36, 1965), Cohen and Rothblum (Linear Algebra Appl. 190, 1993), Gassiat and Boucheron (IEEE Trans. Inform. Theory 49, 2003), Vorob'ev (Theory Probab. Appl. 7, 1962), Kellerer (Z. Wahrsch. 3, 1964) and Arnold, Castillo and Sarabia (Springer, 1999). I updated the companion citations: the foundations paper under its new title, The Physics of the Instant, with its PhilSci-Archive earlier version, and the Koopman, convex-perturbation and selection-functional papers with their current revision numbers. Every anchor the companion papers cite keeps its number: Proposition 4.6, Lemma 4.8, Example 5.2, Proposition 5.3, Section 6 and Proposition 7.1; the new environments are appended within their sections.

Let me know your thoughts.

Ben

Benjamin D. Summers
Managing Director
ADAGIO GROUP
MOBILE +1.713.775.1011 | **OFFICE** +1.305.433.1823 | **FAX** +1 239.932.5558
bsummers@adagiollc.com | adagio.capital

2 attachments

Paper 5 - Memory Rank EB Review.pdf
283K

Paper 5 - Memory Rank SI Review.pdf
171K